

Zahidul Islam Laku

Ph.D. Student, Penn State

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Energy and Mineral Engineering

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Research Focus

My research focuses on a fundamental challenge in fuel science: understanding how trapped gases evolve in porous materials under multiphase systems. Control and prediction of gas redistribution at the pore scale is critical to the reliability of underground hydrogen storage, geologic CO₂ sequestration, and the efficiency of fuel cells and electrolyzers—technologies central to clean energy storage and conversion. To this end, I integrate computational modeling with microfluidic experimentation to develop predictive frameworks that connect microscale physics to macroscopic observations.

Education

Ph.D. | Pennsylvania State University

Fall 2024–Present

Energy and Mineral Engineering | GPA: 3.91/4.00

Advisor: Yashar Mehmani

B.Sc. | Bangladesh University of Engineering & Technology (BUET)

Naval Architecture and Marine Engineering

Work Experience

Research:

Pennsylvania State University | Energy and Mineral Engineering

Fall 2024–Present

Graduate Research Assistant

– *Pore-Scale Modeling of Ostwald Ripening in porous media.*

Bangladesh University of Engineering & Technology | Naval Arch. and Marine Engineering

Undergraduate Research Assistant

– *Numerical investigation of hydrodynamic performance of conventional and ducted propellers.*

Teaching:

Pennsylvania State University | Energy and Mineral Engineering

Graduate Teaching Assistant

– *Reservoir Modeling (PNG 430) [Fall'24].*

– *Special Topics (PNG 497) [Spring'25].*

– *Subsurface Storage Engineering (PNG 460) [Spring'26].*

Brac University | Mathematics and Natural Sciences

Teaching Assistant / Lab Instructor

– *Game Theory (MAT 101) [Teaching Assistant- Spring'23].*

– *Integral Calculus and Differential Equations (MAT 101) [Lab Instructor- Fall'23, Spring'24].*

Industry:

Softeko | Dhaka, Bangladesh

Content Development Executive [Summer'23, Summer'24]

Developed technical documentation and instructional content for the LinuxSimply project, focusing on Linux fundamentals, command-line tools, and system administration topics.

Chittagong Dry Dock | Chittagong, Bangladesh

Engineer Intern [Spring'20]

Assisted in vessel docking and undocking operations, supported hull inspection and surface preparation for marine coating and painting, and participated in routine maintenance and repair activities to ensure operational safety and structural integrity.

Publications

1. Laku, M. Z. I., Salehpour, M., Lan, T., Zhao, B., and Mehmani, Y., "Modeling Ostwald Ripening Dynamics in Porous Microstructures." *arXiv preprint* arXiv:2604.06581 (2026); (submitted to *Journal of Computational Physics*).
2. Salehpour, M., Lan, T., Bueno, N., Laku, M. Z. I., Mehmani, Y., and Zhao, B., "Ostwald Ripening in Underground Gas Storage." *Geophysical Research Letters*, 53, e2025GL120691 (2026).
3. Tarafder, M. S., Haque, M. I., Asaduzzaman, M., & Laku, M. Z. I., "Numerical investigation of hydrodynamic performance of conventional and ducted propellers." *Journal of Naval Architecture and Marine Engineering* (2023).

Poster Presentations

- Predictive Modeling of Ostwald Ripening in Porous Microstructures, AGU Fall Meeting, 2025.
- Image-based pore-network modeling of Ostwald Ripening, EME Student Poster Contest, 2025.

Community Service

- Treasurer, Bangladesh Student Association, Penn State (2025–2026 Executive Committee).
- Judge, Pennsylvania Junior Academy of Science (PJAS), 2026.

Honors and Awards

- EME Outstanding Graduate Teaching Assistant Award (2025), Pennsylvania State University.
- Serena DiMagno Memorial Scholarship (2026), Water Works Operators' Association of Pennsylvania (WWOAP).
- University Merit Award (2017 & 2018), Bangladesh University of Engineering & Technology (BUET).

Skills

- **Computer Programming:** MATLAB, Python, C/C++
- **Miscellaneous:** L^AT_EX, Microsoft Office, Google Workspace, AutoCAD
- **Operating Systems:** Linux, Windows, macOS

Languages

- English (fluent); Bengali (fluent); Hindi (conversational); Urdu (conversational); Spanish (learning).